



GUI Development

Practical Skills Module 5: Topic 2

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Topic Code	PM-05-PS02
Topic	Get started with GUI platform

Scope of Practical Skill

Given a range the learner must be able to:

- PA0201 Import selected module
- PA0202 Create the GUI application main window
- PA0203 Add widgets to the GUI application
- PA0204 Handle user interactions with the event loop
- PA0205 Structure the GUI layout

Applied Knowledge

- AK0201 Concept, definition and functions of GUI

Internal Assessment Criteria

- IAC0201 Expected results with Python GUI are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-05



Practical task

Follow the facilitator's instructions to complete the following activities:

A **student management application** allows users to input and display student data (like name and age) through a graphical user interface (GUI). The application uses Python's PyQt library to create the GUI, manage user input, and respond to events like button clicks.

Create a Python file called **student_management.py** and complete the following:

Step 1: Import selected module

- Import the **PyQt5.QtWidgets** module to access the necessary classes for building the GUI.

Step 2: Create the GUI application main window

- Create the main window for the student management application.
- **QApplication** can be used to initialize the app and **QWidget** can be used to create the main window.

Step 3: Add widgets to the GUI application

- Add input fields, labels, and buttons to collect and display student information.
- You can use widgets such as **QLabel** (for labels), **QLineEdit** (for text input), and **QPushButton** (for user actions).

Step 4: Handle user interactions with the event loop

- Respond to user actions like button clicks to save and display student data.
- You can set up this loop using **app.exec_()** to make the app interactive.

Step 5: Structure the GUI layout

- Combine all widgets to create a complete GUI for managing student data.

- You can use layout managers like **QVBoxLayout** or **QGridLayout** to arrange your widgets neatly on the main window.



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
